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COMMON HALF YEARLY EXAMINATION - 2025

Standard XII

Reg.No. | | | | | | |

PHYSICS

Time : 3.00 hrs **Part – I** *Marks : 70*

I. Choose the correct answer: **15 x 1 = 15**

1. An em wave is propagating in a medium with a velocity $\vec{v} = v\hat{i}$. The instantaneous oscillating electric field of this em wave is along +y axis, then the direction of oscillating magnetic field of the em wave will be along
   a) –y direction b) –x direction c) +z direction d) –z direction
2. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is
   a) 0.2 Ω b) 0.5 Ω c) 0.8 Ω d) 1.0 Ω
3. In an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is
   a) $Q/2$ b) $Q/\sqrt{3}$ c) $Q/\sqrt{2}$ d) Q
4. Two points A and B are maintained at a potential of 7V and –4V respectively. The work done in moving 50 electrons from A to B is,
   a) 8.80×10^{-17} J b) -8.80×10^{-17} J c) 4.40×10^{-17} J d) 5.80×10^{-17} J
5. The magnetic field at the centre 'O' of the following current loop is



   a) $\frac{\mu_0 I}{4r} \otimes$ b) $\frac{\mu_0 I}{4r} \odot$ c) $\frac{\mu_0 I}{2r} \otimes$ d) $\frac{\mu_0 I}{2r} \odot$
6. If the mean wavelength of light from Sun is taken as 550 nm and its mean power as 3.8×10^{26} w, then the number of photons emitted per second from the Sun is of the order of
   a) 10^{45} b) 10^{42} c) 10^{54} d) 10^{51}
7. The frequency range of 3 MHz to 30 MHz is used for
   a) Ground wave propagation b) Space wave propagation
   c) Sky wave propagation d) Satellite communication
8. The radius of curvature of curved surface at a thin plano convex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be
   a) 5 cm b) 10 cm c) 15 cm d) 20 cm
9. In a hydrogen atom, the electron revolving in the fourth orbit, has angular momentum equal to
   a) h b) h/π c) $4h/\pi$ d) $2h/\pi$
10. The transverse nature of light is shown in
    a) interference b) diffraction c) scattering d) polarisation
11. If the input to the NOT gate is A = 1101 its out put is
    a) 0100 b) 1000 c) 0010 d) 0011
12. The gravitational waves were theoreticall proposed by
    a) Conrad Rontgen b) Marie Curie c) Albert Einstein d) Edward Purcell

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13. The time lag between the illumination of surface and photo electric emission is
a) less than 10^{-9} s b) greater than 10^{-9} s c) less than 10^{9} s d) greater than 10^{9} s
14. The emprical relation connecting raduis of nucleus (Z > 10) R and mass number A is given by $R = R_0 A^{1/3}$ where R_0 = _____.
a) 1.2 m b) 2.1 m c) 2.1 F d) 1.2 F
15. The Di–electric strength of air is
a) 0.3×10^6 vm^{-1} b) 14×10^6 vm^{-1} c) 3×10^6 vm^{-1} d) 16×10^6 vm^{-1}

Part - II

II. Answer any 6 questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Two materials X and Y are magnetised whose values of intensity of magnetisation are 500 Am^{-1} and 2000 Am^{-1} respectively. If the magnetising field is 1000 Am^{-1}, then which one among these material can be easily magnetised.
17. Write a note on electro-stalic shielding.
18. Why capacitor blocks DC ?
19. State Malus's law.
20. What is optical path?
21. What is Peltier effect?
22. What are black holes?
23. Define threshold frequency.
24. The radius of the 5th orbit of hydrogen atom is 13.25 A°. Calculate the de–broglie wavelength of the electron orbitting in the 5th orbit.

Part - III

III. Answer any 6 questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Write down the properties of electromagnetic waves.
26. Discuss about Nicol prism.
27. List out the advantages and limitations of Amplitude Modulation.
28. Obtain the expression for energy stored in the parallel plate capacitor.
29. State and explain Biot-Savert law.
30. How do we obtain characteristic X–ray spectra.
31. How will you induce emf by changing the area enclosed by the coil?
32. Obtain the equation for apparent depth.
33. Find the heat energy produced in a resistance of 10Ω when 5 A current flows through it for 5 minutes.

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) Explain the construction and working of a transformer. ***(OR)***
b) Derive the energy expression for an electron in the hydrogen atom using Bohr atom model.
35. a) Derive an expression for electrostatic potential due to an electric dipole. ***(OR)***
b) Obtain lens make's formula and mention its significance.
36. a) Obtain a relation for the magnetic field at a point along the axis of a circulor coil carrying current using Biot Savart law. ***(OR)***
b) Describe briefly Davisson–Germer experiment which demonstrated the wave nature of electrons.
37. a) Explain the determination of unknown resistance using meter bridge. ***(OR)***
b) Explain about compound microscope and obtain the equation for the magnification
38. a) Write down Maxwell's equations in integral form. ***(OR)***
b) Explain the construction and working of a full-wave rectifier.

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